

MacroMap

A grounded Financial Analyst Copilot using prompt engineering, Retrieval-Augmented Generation (RAG), verified computation, and optional lightweight fine-tuning.

Prepared by	Ruthwik Dovala
Date	January 21, 2026
Project type	Full-stack portfolio system (cost-aware, production-minded)

1. Executive Summary

MacroMap is a financial analysis assistant that answers analytical questions with evidence-backed citations and verified computations. It combines (1) prompt engineering for structured analyst-grade outputs, (2) RAG over SEC filings and curated datasets, (3) a computation layer for ratios and time-series analytics, and (4) optional low-cost LoRA fine-tuning to improve tool-use consistency and formatting.

Primary outcome: a credible, demo-ready full-stack application that produces accurate, traceable analysis memos, tables, and charts at low operating cost.

2. Problem Statement

Many finance chatbots produce fluent narratives but fail on three critical requirements for real analysis: (a) traceability to primary sources, (b) numeric correctness, and (c) handling of data freshness. MacroMap addresses these gaps by forcing evidence retrieval for claims, using tools for computations, and clearly separating sourced facts from model-generated reasoning.

3. Objectives

- Answer finance questions with citations to exact filing sections and timestamps/locations.
- Compute key metrics (margins, growth, liquidity ratios, leverage, cash flow metrics) from structured facts rather than guessing.
- Support cross-company comparisons using consistent metric definitions and normalization rules.
- Generate reusable outputs: analyst memo, KPI table, and chart pack for a selected company/time window.
- Operate under cost constraints through small-model inference, caching, and selective escalation.

4. Scope and Users

- Users: students, early-career analysts, founders, and anyone doing fundamentals-driven company research.
- Supported question types: trend analysis (YoY/QoQ), driver attribution (what changed and why), risk-factor change summaries, and macro context integration.
- Initial universe: U.S.-listed companies with SEC filings; extensible to other sources later.

5. Data Sources

- SEC EDGAR filings (10-K, 10-Q, 8-K) for narrative evidence and disclosure context.
- XBRL facts (structured fundamentals) for numeric extraction and reproducible computations.
- Macro indicators (e.g., rates, CPI, unemployment) for contextual overlays and simple correlations.
- Optional: user-uploaded PDFs or CSVs (internal research notes or custom datasets).

6. Solution Approach and Architecture

MacroMap uses a three-layer reasoning design: Retrieval -> Tools -> Synthesis. The system retrieves relevant passages and structured facts, runs explicit computations, and then writes a grounded response in a strict template (answer, evidence, calculations, assumptions, limitations).

Layer	Key responsibilities	Cost-control tactics
Ingestion	Fetch filings and XBRL; clean HTML; section-aware chunking; metadata tagging	Incremental updates; store raw + parsed artifacts once
Retrieval (RAG)	Vector search with filters (company, form, date, section); return evidence chunks	Small embeddings; caching; constrained context windows
Tools/Compute	Ratios, growth, time-series pulls, chart generation, comparisons	Deterministic compute; reuse cached results
LLM Orchestrator	Prompted synthesis + tool routing; refusal when evidence missing	Route easy queries to small model; escalate selectively
App/API	Chat UI, citations panel, memo export, KPI dashboard	Server-side rendering; rate limits; request batching

7. Fine-tuning Strategy (Optional, Budget-Safe)

Fine-tuning is deliberately deferred until the RAG + tool pipeline is stable. If applied, MacroMap uses LoRA/QLoRA on a compact instruction model to improve: (1) citation discipline, (2) tool selection reliability, and (3) consistent memo formatting. Training data will be a curated set of high-quality question -> tool traces -> gold answers, emphasizing correctness over scale.

8. Evaluation and Quality Targets

- Groundedness: every substantive claim is supported by retrieved evidence or explicit computations.
- Numeric accuracy: computed outputs match deterministic tool results (unit-tested).
- Citation precision: citations point to the correct filing section and match the claim they support.
- Robustness: prompt-injection resistance when filings contain misleading instructions.
- Latency/cost: predictable per-query cost via caching and model routing.

9. Deliverables

- Full-stack web application (chat + company workspace) with a clean UI (authentication optional).
- Backend API for ingestion, retrieval, tool execution, and orchestration.
- Evaluation harness with a reproducible benchmark set and scoring report.
- Demo dataset and scripted walkthroughs (common analyst workflows).
- Documentation: architecture diagram, data lineage notes, deployment guide.

10. Implementation Timeline (6-week plan)

Week	Milestones
1	EDGAR ingestion, parsing, and section-aware chunking; initial storage schema
2	RAG retrieval with citations; baseline chat experience; caching
3	XBRL fact extraction + ratio engine; KPI table outputs
4	Macro integration + chart tool; cross-company compare view
5	Evaluation harness + test set; hardening (security, injection, edge cases)
6	Optional LoRA fine-tune; before/after evaluation; final polish + demo

11. Budget and Cost Controls

- Prefer small open-weight models for most queries; escalate only when confidence or retrieval quality is low.
- Aggressive caching: embeddings, retrieval results, and computed metrics keyed by filing version.
- Limit context size by section-based retrieval and strict response templates.
- Batch ingestion and incremental updates to avoid repeated parsing/embedding.
- Run training/inference on low-cost options (local GPU or short-lived GPU instances) when feasible.

12. Risks and Mitigations

- Hallucinated numbers: mitigated by tool-first numeric policy and unit tests for computations.
- Incorrect citations: mitigated by citation validation (claim-to-chunk matching checks) and evaluation scoring.
- Prompt injection from documents: mitigated by system prompt hardening and retrieval sanitization.
- Data gaps or non-standard filings: mitigated by graceful fallbacks and transparent 'missing data' responses.